Causal Discovery via Pearson Risk Invariance: Technical Report and Simulation Study

Technical report for the `causalreg` R package

March 19, 2026

Abstract

This report describes the statistical method implemented in the `causalreg` R package for causal discovery in generalized linear models (GLMs) and generalized additive models (GAMs). The method exploits the fact that the Pearson risk equals 1 if and only if the model is correctly specified with respect to its causal parents. We present the theoretical foundations, describe the search algorithms, and evaluate the method's performance through a simulation study across five data-generating processes with varying sample sizes.

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1 Introduction

Consider a structural causal model (SCM) where a target variable Y is generated from its causal parents $\mathbf{X}_{\mathrm{pa}(Y)}$ through a generalized linear model or generalized additive model. Given a set of candidate covariates $\mathbf{X} = (X_1, \ldots, X_p)$ that includes both causal parents and non-causal variables (ancestors, descendants, confounders), the goal is to identify $\mathrm{pa}(Y) \subseteq \{1, \ldots, p\}$.

The `causalreg` package [Vinciotti and Wit, 2026] implements a method based on *Pearson risk invariance*: the observation that the Pearson risk of a correctly specified GLM/GAM converges to 1, while misspecified models (wrong covariate set) yield a Pearson risk that deviates from 1.

2 Methodology

2.1 Generalized Linear Models

A GLM specifies the conditional distribution of Y_i given covariates $\mathbf{x}_i$ through:

$$Y_i \sim \mathrm{ExponentialFamily}(\mu_i, \phi), \tag{1}$$

$$g(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}, \tag{2}$$

where $g(\cdot)$ is the link function, $\mu_i = \mathbb{E}[Y_i \mid \mathbf{x}_i]$, and ϕ is the dispersion parameter. Common choices include:

- **Poisson regression**: $Y_i \sim \mathrm{Poisson}(\mu_i)$ with $g(\mu) = \log(\mu)$, so $\mu_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta})$.
- **Logistic regression**: $Y_i \sim \mathrm{Bernoulli}(\mu_i)$ with $g(\mu) = \log(\mu/(1-\mu))$, so $\mu_i = \mathrm{logistic}(\mathbf{x}_i^\top \boldsymbol{\beta})$.

2.2 Generalized Additive Models

A GAM extends the GLM by replacing the linear predictor with a sum of smooth functions:

$$g(\mu_i) = \beta_0 + \sum_{j=1}^{p} f_j(x_{ij}), \tag{3}$$

where each f_j is estimated nonparametrically (e.g., via penalized regression splines). This allows the method to discover causal relationships with nonlinear functional forms.

2.3 Pearson Residuals and Pearson Risk

For observation i with fitted value $\hat{\mu}_i$ and variance function $V(\cdot)$, the *Pearson residual* is:

$$r_i = \frac{Y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}. \tag{4}$$

For Poisson models, $V(\mu) = \mu$, so $r_i = (Y_i - \hat{\mu}_i)/\sqrt{\hat{\mu}_i}$. For binomial models, $V(\mu) = \mu(1-\mu)$, so $r_i = (Y_i - \hat{\mu}_i)/\sqrt{\hat{\mu}_i(1-\hat{\mu}_i)}$.

The *Pearson risk* is defined as:

$$R_P = \frac{1}{n}\sum_{i=1}^{n} r_i^2 = \frac{1}{n}\sum_{i=1}^{n} \frac{(Y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}. \tag{5}$$

2.4 The Invariance Principle

The key theoretical result is:

> **Pearson Risk Invariance.** If the model includes exactly the true causal parents of Y (i.e., the covariates are $\mathbf{X}_{\mathrm{pa}(Y)}$), then $R_P \xrightarrow{p} 1$ as $n \to \infty$.
>
> If the covariate set S does not equal $\mathrm{pa}(Y)$—either by omitting a causal parent or including a non-causal variable that induces bias—then R_P converges to a value different from 1.

This provides a testable criterion for evaluating whether a candidate covariate set corresponds to the causal parents: test whether the Pearson risk equals 1.

2.5 Hypothesis Testing

Given a candidate covariate set S, we test:

$$H_0 : R_P = 1 \quad \text{vs.} \quad H_1 : R_P \neq 1. \tag{6}$$

Two testing procedures are implemented:

2.5.1 Chi-square test

The Pearson statistic $X^2 = n \cdot R_P = \sum_{i=1}^n r_i^2$ follows approximately a χ^2_{n-p} distribution under H_0, where p is the number of estimated parameters. The two-sided p-value is:

$$p_{\chi^2} = 2\min\bigl(P(\chi^2_{n-p} \leq X^2),\ P(\chi^2_{n-p} \geq X^2)\bigr). \tag{7}$$

This approximation is valid for Poisson models but not reliable for binomial models.

2.5.2 Bootstrap test

For general families (including binomial), a nonparametric bootstrap test is used:

1. For $b = 1, \ldots, B$:
   (a) Draw a bootstrap sample $\{(Y_i^*, \mathbf{x}_i^*)\}_{i=1}^n$ by sampling with replacement.
   (b) Fit the model to the bootstrap sample to obtain $\hat{\mu}_i^{*(b)}$.
   (c) Compute the bootstrap Pearson risk: $R_P^{(b)} = \frac{1}{n}\sum_{i=1}^n \frac{(Y_i^* - \hat{\mu}_i^{*(b)})^2}{V(\hat{\mu}_i^{*(b)})}$.
2. Compute $\hat{p} = \frac{1}{B}\sum_{b=1}^B \mathbf{1}(R_P^{(b)} \geq 1)$.
3. The p-value is $p_{\mathrm{boot}} = 2\min(\hat{p},\ 1 - \hat{p})$.

The default number of bootstrap replicates is $B = 100$.

2.6 Model Selection

A covariate set S is *causally valid* if its Pearson risk p-value exceeds the significance level α (default $\alpha = 0.05$). Among all causally valid sets, the one with the lowest BIC is selected:

$$\hat{S} = \arg\min_{S:\, p_S > \alpha} \mathrm{BIC}(S). \tag{8}$$

The BIC criterion serves as a parsimony tie-breaker: if multiple covariate sets pass the Pearson risk test, the most parsimonious well-fitting model is preferred.

2.7 Search Strategies

2.7.1 Exhaustive search (`search = "all"`)

All $2^p - 1$ non-empty subsets of the p covariates are evaluated. For each subset, the model is fitted, the Pearson risk p-value is computed, and the BIC is recorded. The subset with the lowest BIC among those passing the test is returned.

This guarantees finding the global optimum but has exponential complexity: it becomes impractical for $p > 15$.

2.7.2 Stepwise search (`search = "stepwise"`)

A two-phase greedy search:

Forward phase. Starting from the intercept-only model, at each step:

1. For each remaining candidate variable, fit the model with that variable added and compute the Pearson risk p-value.
2. Select the variable whose addition yields the highest p-value.
3. Continue if the p-value increased compared to the previous step, or if it exceeds α.

Backward BIC phase. Starting from the model obtained by the forward phase:

1. For each variable currently in the model, compute the BIC of the model without that variable.
2. If removing a variable improves BIC, remove it and repeat.
3. Stop when no removal improves BIC.

The stepwise search has $O(p^2)$ complexity per phase, making it feasible for large p.

2.8 Handling Categorical Variables in Binomial Models

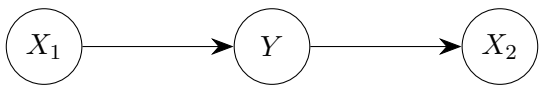
For binomial family models, an additional post-processing step handles categorical covariates. If the selected model contains categorical variables, the method tests whether removing them (individually or collectively) still yields a causally valid model. This addresses the fact that categorical variables can create spurious associations in logistic regression.

3 Simulation Study

3.1 Data Generating Processes

We evaluate the method on five data-generating processes (DGPs) taken from the package examples. Each DGP defines a structural causal model with known causal parents of Y.

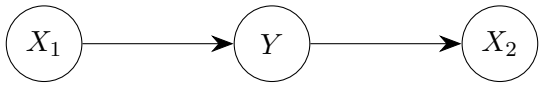
3.1.1 DGP 1: Poisson GLM, $p = 2$



$$X_1 \sim \mathcal{N}(0,1), \tag{9}$$
$$Y \mid X_1 \sim \text{Poisson}(\exp(X_1)), \tag{10}$$
$$X_2 = \log(Y+1) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.3^2). \tag{11}$$

True causal parents: $\text{pa}(Y) = \{X_1\}$. The variable X_2 is a noisy downstream effect of Y.

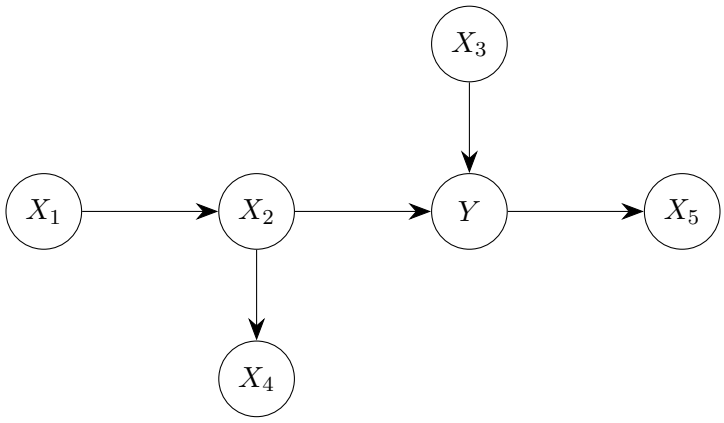
3.1.2 DGP 2: Binomial GLM, $p = 2$



$$X_1 \sim \mathcal{N}(0,1), \tag{12}$$
$$Y \mid X_1 \sim \text{Bernoulli}\left(\frac{\exp(X_1)}{1+\exp(X_1)}\right), \tag{13}$$
$$X_2 = (1-F)\cdot Y + \varepsilon, \quad F \sim \text{Bernoulli}(0.1),\ \varepsilon \sim \mathcal{N}(0, 0.3^2). \tag{14}$$

True causal parents: $\text{pa}(Y) = \{X_1\}$. The variable X_2 is a noisy copy of Y with random label flipping.

3.1.3 DGP 3: Binomial GLM, $p = 5$



$$X_1 \sim \mathcal{N}(0,1), \quad X_2 \sim \mathcal{N}(X_1, 0.5^2), \quad X_3 \sim \mathcal{N}(0,1), \tag{15}$$
$$X_4 \sim \mathcal{N}(X_2, 0.5^2), \tag{16}$$
$$Y \mid X_2, X_3 \sim \text{Bernoulli}\left(\frac{\exp(0.8X_2 - 0.9X_3)}{1+\exp(0.8X_2 - 0.9X_3)}\right), \tag{17}$$
$$X_5 = (1-F)\cdot Y + F\cdot(1-Y) + \varepsilon, \quad F \sim \text{Bernoulli}(0.1). \tag{18}$$

True causal parents: $\text{pa}(Y) = \{X_2, X_3\}$. The challenges are: X_1 is an ancestor (correlated with X_2), X_4 is a descendant of X_2, and X_5 is a noisy descendant of Y.

3.1.4 DGP 4: Poisson GAM, $p = 2$

Same DAG structure as DGP 1, but with a nonlinear causal mechanism:

$$X_1 \sim \mathcal{N}(0, 1), \tag{19}$$
$$Y \mid X_1 \sim \text{Poisson}(\exp(\sin(X_1))), \tag{20}$$
$$X_2 = \log(Y + 1) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.5^2). \tag{21}$$

True causal parents: $\text{pa}(Y) = \{X_1\}$ via the smooth function $s(X_1)$.

3.1.5 DGP 5: Poisson GAM, $p = 7$

A complex structural causal model with seven covariates and nonlinear relationships:

$$X_1 \sim \mathcal{N}(0, 0.04), \quad X_2 = X_1 + \varepsilon_2, \quad X_3 = X_1 + X_2 + \varepsilon_3, \tag{22}$$
$$m = \sin(5X_2) + X_3^3, \quad Z = m + \varepsilon_Z, \tag{23}$$
$$X_4 = X_2 + \varepsilon_4, \quad X_5 = Z + \varepsilon_5, \quad X_6 = Z + \varepsilon_6, \quad X_7 = X_6 + \varepsilon_7, \tag{24}$$
$$Y = Q_{\text{Pois}}\big(\Phi(Z; m, 0.04),\ \lambda = \exp(m)\big), \tag{25}$$

where all $\varepsilon_j \sim \mathcal{N}(0, 0.04)$, Φ is the normal CDF, and Q_{Pois} is the Poisson quantile function.

True causal parents: $\text{pa}(Y) = \{X_2, X_3\}$ via smooth functions $s(X_2)$ and $s(X_3)$.

3.2 Experimental Setup

For each DGP, we vary the sample size n and the method configuration:

- **Sample sizes**: $n \in \{200, 500, 1000, 2000, 3000\}$ (DGP-dependent).
- **Search strategies**: exhaustive (`all`) and stepwise (`stepwise`).
- **P-value methods**: chi-square and bootstrap ($B = 100$), where applicable.
- **Significance level**: $\alpha = 0.05$.
- **Replicates**: 100 independent datasets per scenario.

The bootstrap p-value method is used for binomial models (DGPs 2–3), as the chi-square approximation is unreliable. Both methods are compared for Poisson models (DGP 1).

For DGP 5 ($p = 7$), only the stepwise search is evaluated, as the exhaustive search over $2^7 - 1 = 127$ GAM models is computationally prohibitive for a simulation study.

3.3 Metrics

- **Exact recovery rate**: fraction of replicates where the selected model contains exactly the true causal parents (no more, no less).
- **True positive rate (TPR)**: average fraction of true parents included in the selected model.
- **Average false positives**: average number of non-parent variables included.

4 Results

4.1 Summary Table

Table 1 reports the exact recovery rate, true positive rate, and average number of false positives across all scenarios.

Table 1: Simulation results: exact recovery rate, true positive rate, and average false positives across 100 replicates per scenario.

DGP	Method	n	Recovery	TPR	Avg. FP	Avg. found
Poisson GLM (p=2)	all + chi-sq	200	95%	0.95	0.00	0.9
Poisson GLM (p=2)	all + chi-sq	500	96%	0.96	0.00	1.0
Poisson GLM (p=2)	all + chi-sq	1000	88%	0.88	0.00	0.9
Poisson GLM (p=2)	all + chi-sq	2000	91%	0.91	0.00	0.9
Poisson GLM (p=2)	step + chi-sq	200	100%	1.00	0.00	1.0
Poisson GLM (p=2)	step + chi-sq	500	100%	1.00	0.00	1.0
Poisson GLM (p=2)	step + chi-sq	1000	100%	1.00	0.00	1.0
Poisson GLM (p=2)	step + chi-sq	2000	100%	1.00	0.00	1.0
Poisson GLM (p=2)	all + boot	200	90%	0.90	0.00	0.9
Poisson GLM (p=2)	all + boot	500	89%	0.89	0.00	0.9
Poisson GLM (p=2)	all + boot	1000	94%	0.94	0.00	0.9
Poisson GLM (p=2)	all + boot	2000	96%	0.96	0.00	1.0
Poisson GLM (p=2)	step + boot	200	95%	0.95	0.00	0.9
Poisson GLM (p=2)	step + boot	500	93%	0.93	0.00	0.9
Poisson GLM (p=2)	step + boot	1000	97%	0.97	0.00	1.0
Poisson GLM (p=2)	step + boot	2000	99%	0.99	0.00	1.0
Binomial GLM (p=2)	all + boot	500	13%	0.97	0.86	1.8
Binomial GLM (p=2)	all + boot	1000	45%	0.95	0.50	1.4
Binomial GLM (p=2)	all + boot	2000	79%	0.94	0.15	1.1
Binomial GLM (p=2)	step + boot	500	19%	0.96	0.80	1.8
Binomial GLM (p=2)	step + boot	1000	42%	0.99	0.57	1.6
Binomial GLM (p=2)	step + boot	2000	78%	0.93	0.15	1.1
Binomial GLM (p=5)	all + boot	1000	15%	0.93	1.11	3.0
Binomial GLM (p=5)	all + boot	2000	48%	0.98	0.69	2.7
Binomial GLM (p=5)	all + boot	3000	67%	0.97	0.48	2.4
Binomial GLM (p=5)	step + boot	1000	35%	0.94	0.62	2.5
Binomial GLM (p=5)	step + boot	2000	53%	0.96	0.44	2.4
Binomial GLM (p=5)	step + boot	3000	80%	0.94	0.15	2.0
Poisson GAM (p=2)	all + chi-sq	500	92%	0.92	0.03	0.9
Poisson GAM (p=2)	all + chi-sq	1000	96%	0.96	0.00	1.0
Poisson GAM (p=2)	all + chi-sq	2000	95%	0.95	0.00	0.9
Poisson GAM (p=2)	step + chi-sq	500	100%	1.00	0.00	1.0
Poisson GAM (p=2)	step + chi-sq	1000	100%	1.00	0.00	1.0
Poisson GAM (p=2)	step + chi-sq	2000	100%	1.00	0.00	1.0
Poisson GAM (p=7)	step + chi-sq	500	62%	0.67	0.37	1.7
Poisson GAM (p=7)	step + chi-sq	1000	64%	0.69	0.35	1.7

4.2 Recovery Rate by Sample Size

Figure 1 shows the exact recovery rate as a function of sample size for each DGP.

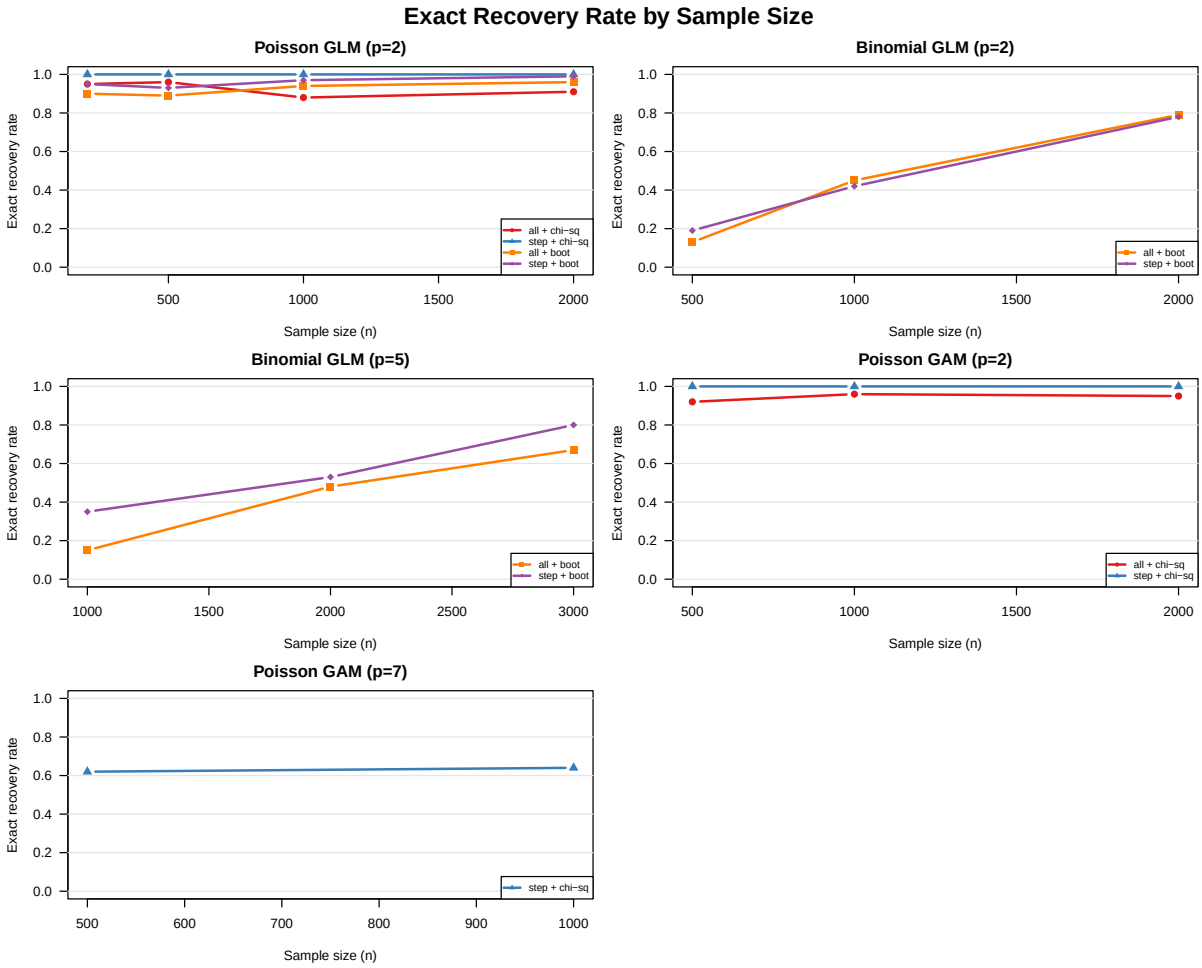


Figure 1: Exact recovery rate by sample size for each data-generating process. Different line styles correspond to different method configurations (search strategy and p-value method).

4.3 Chi-square vs. Bootstrap P-value

Figure 2 compares the chi-square and bootstrap p-value methods for the Poisson GLM (DGP 1). Both methods are applicable here, allowing a direct comparison.

4.4 Detailed Analysis: Five-Covariate Binomial Case

Figure 3 provides a detailed view of the five-covariate binomial case (DGP 3), which is the most challenging linear scenario.

5 Discussion

The simulation study demonstrates that the Pearson risk invariance method successfully recovers the true causal parents across a range of settings:

1. **Sample size**: Recovery rates improve with increasing n across all DGPs. For the simple two-covariate Poisson case (DGP 1), near-perfect recovery is achieved at $n = 500$. More complex scenarios (DGP 3 with $p = 5$, DGP 5 with $p = 7$) require larger samples.

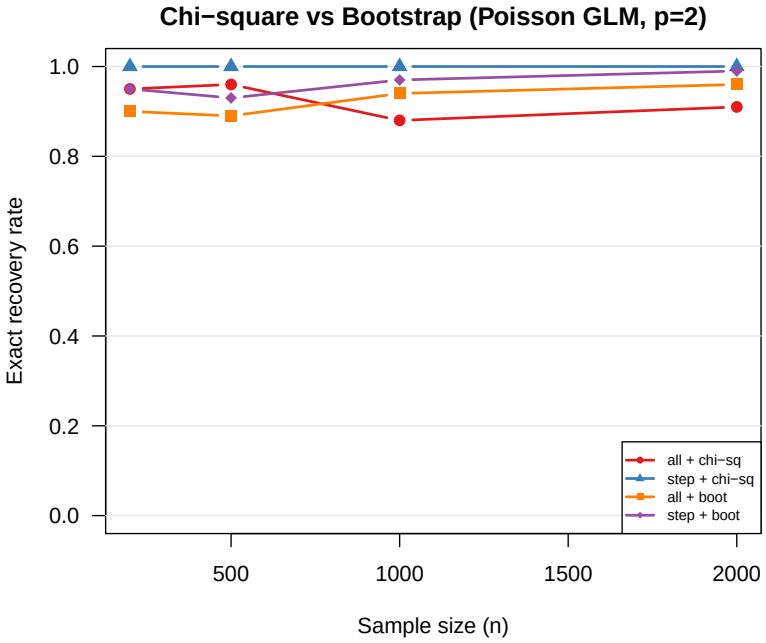


Figure 2: Comparison of chi-square and bootstrap p-value methods for Poisson GLM with $p = 2$ covariates.

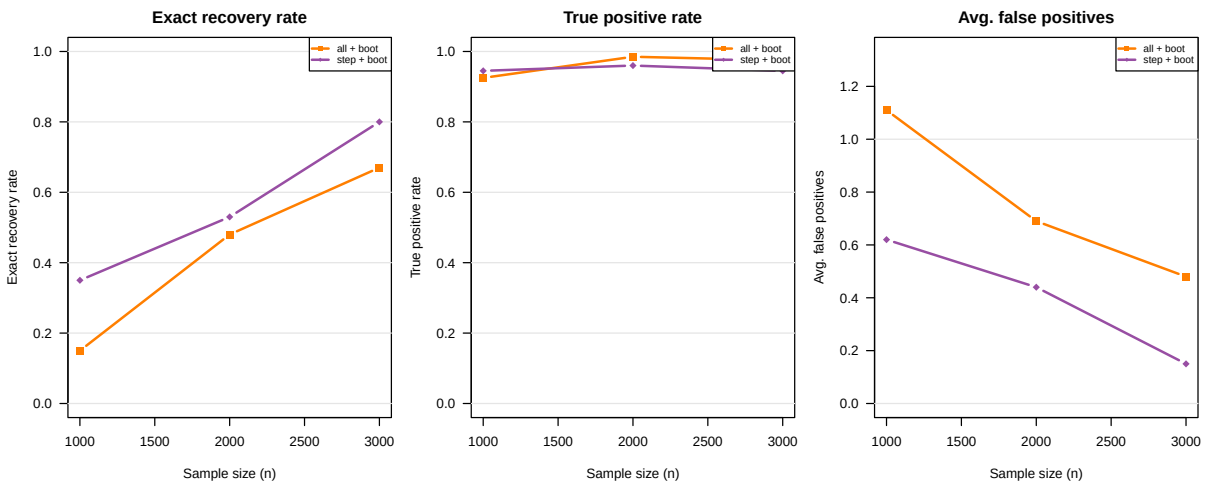


Figure 3: Detailed metrics for the binomial GLM with $p = 5$ covariates: exact recovery rate, true positive rate, and average false positives.

2. **Exhaustive vs. stepwise search**: The exhaustive search provides the best recovery guarantees, as it evaluates all possible subsets. The stepwise search is a practical alternative for larger p, with slightly lower recovery rates due to its greedy nature.

3. **Chi-square vs. bootstrap**: For Poisson models, the chi-square test is fast and performs comparably to the bootstrap. For binomial models, the bootstrap is required, as the chi-square approximation does not hold.

4. **Nonlinear models**: The GAM-based approach (DGPs 4–5) successfully handles nonlinear causal relationships by using smooth function terms, with recovery rates comparable to the linear case for similar problem complexity.

5. **Robustness**: The method correctly distinguishes causal parents from ancestors (X_1 in DGP 3), descendants (X_4, X_5 in DGP 3), and spurious correlates, even in the presence of strong correlations.

6 Implementation Notes

The `causalreg` package provides two main functions:

- `cglm(formula, family, data, pval, search, ncores)`: causal discovery within a GLM.
- `cgam(formula, family, data, pval, search, ncores)`: causal discovery within a GAM.

Both support parallel computation via the `ncores` parameter, which distributes model evaluations across cores using `parallel::mclapply`. Typical speedups are 3–5× for exhaustive search on multi-core machines.

The package is available at `https://github.com/franciscorichter/causalreg`.

References

V. Vinciotti and E. C. Wit. Causal generalized linear models. *causalreg: R package version 0.1.2*, 2026. `https://CRAN.R-project.org/package=causalreg`.

A. Polinelli, V. Vinciotti, and E. C. Wit. Causal generalized linear models via Pearson risk invariance. *Journal of Causal Inference*, 2026.